

Devam Dholakia

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Computer Science — GPA: 3.93

Aug. 2024 – May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Systems Software, Enterprise Computing, Security in Computing, Object-Oriented Software Development, Computer Networks

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

USAA (United Services Automobile Association)

San Antonio, TX

- Collapsed a **4+** screen rep lookup into a single round-trip by resolving a customer's linked-account graph server-side, shipping the feature in **Spring Boot** with a **React/Redux** client.
- Drove end-to-end development of the feature, from data model and API contract through production rollout, targeting the service-rep workflow that drives **average handle time**.
- Migrated **6** legacy internal-service integrations onto **GraphQL**, replacing brittle REST response parsing with a typed contract that lets callers select only the fields they need.

Research Assistant

Apr. 2026 – Present

Interactive Systems and User Experience (ISUE) Lab, UCF

Orlando, FL

- Analyzed and synthesized methodologies from **65** papers to construct the related-works framework for a **SIGGRAPH** submission on language-conditioned 3D scene generation.
- Automated NASA-TLX and SUS survey scoring by engineering a **QLoRA** fine-tuning pipeline for **LLaMA** on consumer-grade hardware, ensembling weighted models to improve accuracy.

PROJECTS

uKnight | *Spring Boot, WebRTC, React, PostgreSQL, Redis, GCP Cloud Run, Docker*

- Led a **6-engineer team** shipping an anonymous video-chat platform to **72** .edu users in **3 weeks**; designed a **Redis**-backed matchmaking queue keeping pairing state out of the app tier so signaling scaled statelessly.
- Kept calls connecting on locked-down campus networks by implementing **WebRTC** signaling over **WebSockets** with **STUN/TURN** fallback, relaying through TURN when symmetric NAT blocks a direct peer path.
- Containerized the signaling and API tiers with **Docker** on **GCP Cloud Run**, gating signup behind .edu verification and persisting account state in **PostgreSQL** so autoscaled instances stayed interchangeable.

Apex — F1 Pit-Wall Simulator | *Next.js, FastAPI, WebSocket, XGBoost, NumPy*

- Engineered a two-stage **Ridge** + **XGBoost** pipeline predicting remaining-race lap times to **0.91s MAE** at mid-race under forward-chaining validation, a **41%** error cut over a mean-of-observed-laps baseline.
- Built a vectorized **NumPy** Monte Carlo engine running **10,000** race simulations at **14ms p95**, streaming strategy updates through **FastAPI** to a **Next.js** client at **120/sec** over **WebSockets**.
- Derived **14** features from **10,244** cleaned laps across **11** Grands Prix and **22** drivers, then traced inflated CV scores to a base-pace target leak and rebuilt validation around grouped and forward-chaining splits.

SAP Archive Decoder — KnightHacks 2025 | *Python, gzip, lz4, zstandard* | *1st Place: Auritas Challenge*

- Decoded **5/5** binary SAP archives to CSV/JSON at **≥95%** field accuracy in **36 hours**, building schema-driven parsers for **7** SAP field types including PACKED/BCD.
- Fingerprinted **gzip**, **lz4**, and **zstandard** container framing per file so one CLI handled every archive variant with no per-file configuration or manual format hints.
- Parallelized per-file decoding with **ProcessPoolExecutor**, adding per-field validators and error-tolerant parsing that quarantines corrupt records and reports them instead of failing the run.

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript (ES6+), SQL

Frameworks: Spring Boot, React, Next.js, Redux, Node.js (Express), FastAPI, GraphQL, WebRTC, WebSockets

ML & Data: XGBoost, scikit-learn, NumPy, pandas, PyTorch (QLoRA/PEFT)

Databases & Infrastructure: PostgreSQL, Redis, Docker, GCP Cloud Run, AWS

Tools & Testing: Git/GitHub, GitHub Actions, JUnit | **Cert:** AWS Certified Cloud Practitioner (CLF-C02)